



KUALA LUMPUR UNIVERSITY OF SCIENCE AND TECHNOLOGY

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Individual Assignment

REVIEW ON DATABASE RELATED JOURNAL ARTICLES AND WHITE PAPERS (TOPIC: DATA MINING & DATA WAREHOUSING)

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1.0 Introduction: Keyword and Abstract

For this individual research assignment, I have focused on the topic of **Data Mining and Data Warehousing**. These two concepts are very important for Data Science students. A Data Warehouse is where we store large amounts of historical data safely. Data Mining is the process of using algorithms to find patterns in that data.

Keywords Used

To find the relevant articles for this review, I used specific keywords on Google Scholar and vendor websites. The keywords I chose were:

- **"Cloud Data Warehouse performance"**: I used this to find articles about how databases work on the internet (cloud) instead of local servers.
- **"Data Mining adoption in SMEs"**: I used this to understand how small businesses are using these tools.
- **"Autonomous Database technology"**: I used this to find the latest white papers about self-driving databases.
- **"Clinical Data Warehousing"**: I used this to find examples of data warehousing in the health sector.

Review of Abstracts

I have selected five articles. Here is a review of their abstracts:

1. **Sousa, Costa, and Santos (2024)**: The abstract of this article explains an experiment. The authors tested two different database structures (schemas) in the cloud. They wanted to see if a "Flat Table" is faster than a "Star Schema." The abstract states that the results depend on the type of question (query) you ask the database.
2. **Al-Adwan et al. (2024)**: This abstract describes a systematic review. The authors looked at many studies to see if Data Mining helps Small and Medium Enterprises (SMEs). The abstract concludes that Data Mining gives companies a "competitive advantage," but it is often too expensive for small companies to adopt.
3. **Smith, Davies, and Johnson (2025)**: The abstract of this medical paper reviews how hospitals use data warehouses. It explains that data warehouses are essential for

seeing the "full picture" of a patient's health. However, the abstract notes that medical data is very complex and current systems need to be more flexible.

4. **Snowflake Inc. (White Paper):** This abstract/executive summary introduces the "Data Cloud." It explains that old data warehouses had a problem: if you wanted more storage, you had to pay for more processing power. Snowflake's abstract claims their new architecture separates these two things, making it cheaper and faster.
5. **Oracle Corp. (White Paper):** This abstract introduces the "Autonomous Database." It describes a database that uses Machine Learning to repair and tune itself. The abstract claims this technology eliminates human error and lowers costs by 80%.

2.0 Literature Review

This section discusses the findings from the five articles in detail. The literature highlights three main themes: Performance, Business Value, and Automation.

2.1 Performance in the Cloud

Moving data to the cloud is a major trend. However, simply moving data does not make it fast. The research by Sousa et al. (2024) shows that the design of the database is still critical. In their experiment, they compared different ways of organizing data. They found that for some tasks, a simple "Flat Table" (denormalized data) was fast. But for complex questions, the traditional "Star Schema" was better. This tells us that even with powerful cloud computers, we still need to structure our data carefully.

2.2 Business and Healthcare Value

Data Mining is useless if it does not help the organization. Al-Adwan et al. (2024) found that for small businesses (SMEs), Data Mining is a tool for survival. It helps them understand what customers want. However, the literature points out a big problem: "Economic Factors." Small companies often cannot afford the software or the experts to run it.

In healthcare, the value is even higher. Smith et al. (2025) explain that "clinical data warehousing" saves lives. By combining data from the pharmacy, the lab, and the doctor, the hospital gets a holistic view. But they also found that technical issues, like different data formats, make this very difficult to do in practice.

2.3 The Rise of Automation

The most exciting finding in the literature is automation. The white papers from Snowflake and Oracle describe a future where the database manages itself.

Snowflake has solved the "concurrency" problem. In the past, if 100 people tried to run a report at the same time, the system would freeze. Snowflake's "multi-cluster" architecture creates new virtual warehouses instantly to handle the load.

Oracle takes this further with the "Autonomous Database." This system patches itself and tunes itself. According to Oracle, this means the database is available 99.995% of the time. It removes the need for a human administrator to do boring maintenance tasks.

3.0 Critical Evaluation

Based on the articles I have read; I have evaluated the current state of Data Mining and Warehousing.

Strengths of the New Technology

The biggest strength I see is the reduction of human error. The Oracle white paper makes a very strong argument that most database failures are caused by humans forgetting to update software or configuring it wrong. By automating this, the data is safer.

Another strength is scalability. Snowflake's approach means that a company can start small and grow very big without changing their software. This is perfect for the startups mentioned in the SME article.

Weaknesses and Gaps

However, I also see weaknesses. The academic article by Sousa et al. (2024) proves that automation is not magic. You cannot just dump data into an autonomous database and expect it to be fast. You still need a Data Architect to decide if you should use a Star Schema or a Flat Table. The white papers do not talk about this; they make it sound too easy.

Another weakness is the **Digital Divide**. The technology described by Oracle and Snowflake is amazing, but the review by Al-Adwan (2024) suggests that small companies are being left behind. They cannot afford these "Enterprise" tools. There is a gap between the advanced technology available and what normal businesses can actually use.

Opinion on Methodology

I also critically evaluated the types of sources. The academic journals (Sousa, Smith) were much more objective. They used experiments and statistics. The white papers (Oracle, Snowflake) were informative but biased. They are trying to sell a product. For example, Oracle focuses heavily on their own hardware (Exadata), which locks the customer into using only Oracle products. As a researcher, it is important to trust the academic results more than the vendor claims.

4.0 Conclusion

In conclusion, this research exercise has given me a deep understanding of Data Mining and Data Warehousing.

I have learned that the technology is shifting from "Manual" to "Autonomous." The days of managing physical servers are ending. Technologies like Snowflake and Oracle Autonomous Database allow us to store infinite amounts of data in the cloud.

However, the main conclusion is that **human skill is still required**. The literature review showed that database design (schema) still affects performance, and that business context is needed to get value from data mining. Automation can handle the boring tasks, but it cannot replace the critical thinking of a Data Scientist.

5.0 Reference List

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